

No-Code Dataset Preparation Platform

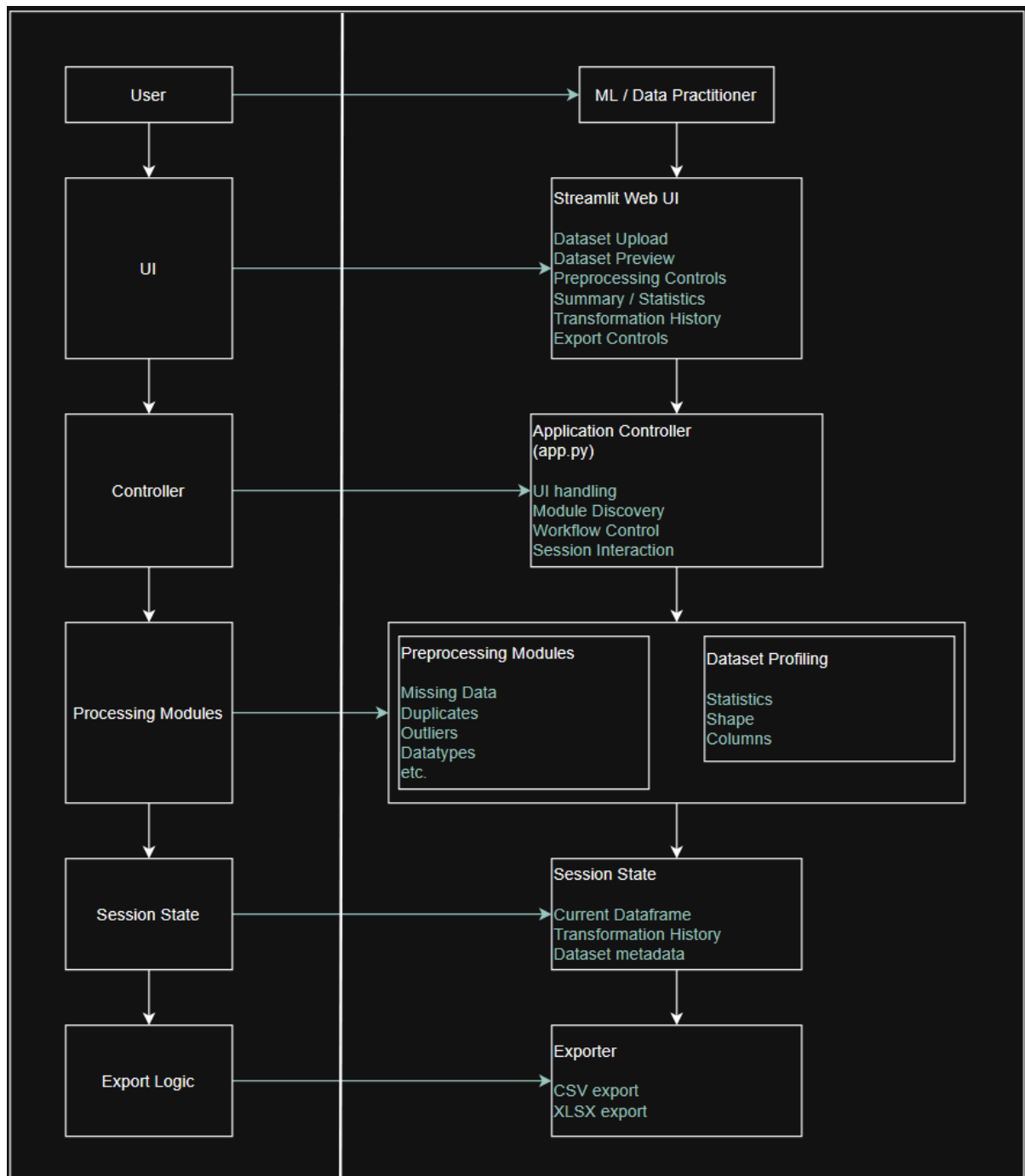
A No Code approach towards data cleaning and preprocessing

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High Level Architecture

The architecture proposed for the project is shown in the diagram below. This is high-level architecture i.e. made from an abstraction point of view. The text points in green show the actions and/or individual modules under the main heading of that specific section



Functional Acceptance Criteria for Current Modules

ID	Requirement	Acceptance Criterion
1	Dataset Upload	System shall successfully accept supported CSV/XLSX datasets
2	Invalid File	System shall reject unsupported or unreadable files with a meaningful error
3	Dataset Preview	System shall display a preview of the uploaded dataset
4	Missing Values	User shall be able to select columns and apply supported missing-value strategies
5	Duplicates	System shall be able to identify and remove duplicate records
6	Data Types	User shall be able to detect and convert supported incorrect data types
7	Outliers	User shall be able to detect and handle supported outlier cases
8	Transformation	System shall update the current dataset after successful preprocessing
9	Export	User shall be able to export the processed dataset
10	Export Integrity	Exported dataset shall contain the transformations performed during the session

Non-Functional Acceptance Criteria for Current Modules

For the MVP, the application should successfully process datasets within the defined development/test dataset limits without crashing and/or freezing.

Reliability

- Invalid files should not crash the application.
- Invalid preprocessing configurations should be rejected.
- Failed transformations should not silently overwrite the dataset.
- The application should provide meaningful error messages.

Usability

- A user without programming knowledge should be able to perform *basic* preprocessing through the UI.
- Every preprocessing operation should clearly state which columns it applies to.
- The UI should prevent incompatible operations from being applied to inappropriate column types.

Maintainability

- Each preprocessing operation should be implemented as an independent module.
- Modules should follow a consistent interface.
- Adding a new preprocessing operation should not require major changes to app.py.
- Modification to the module file and the registry file should be sufficient to implement the module (with minimal changes to app.py).

SDLC Approach

Following table describes the plan to implement SDLC approach in the development and maintenance of the project. A detailed document going over each phase will be produced shortly after approval of the current methodology and scope of the project.

Sr.	SDLC Phase	Project Activity
1	Requirement Analysis	Identify ML/data-preparation problems and user requirements
2	System Design	Architecture, UML diagrams, module design, UI design
3	Implementation	Develop Streamlit application and preprocessing modules
4	Testing	Unit, integration, system and user acceptance testing
5	Deployment	Deploy Streamlit application
6	Maintenance	Fix bugs, improve preprocessing methods, support larger datasets

Target Users

The target users can be defined as ML / Data Science practitioners - people who received raw dataset and need to clean and preprocess it before using it for machine learning or data analysis without wanting to write preprocessing codes manually.

Actor	How they use the Application	Priority
ML Engineer	Performs initial dataset preparation before building ML pipelines	Primary
Data Scientist	Cleans and preprocesses datasets before model development	Primary
Data Analyst	Cleans messy datasets for analysis/visualization	Secondary
ML/DS Student / Researcher	Experiments with preprocessing techniques without writing extensive code	Secondary
System Administrator	Deploys/maintains the application	Supporting, if applicable

For future use cases and generalised explanations, the term “ML / Data Practitioner” will be the most appropriate as it includes all the above-mentioned roles. Alternatively, given the context, the term “User”, too can be used to indicate the above-mentioned roles.